

Automated Revenue Assurance & Demurrage Platform

Commercial Feasibility, Financial Justification & Strategic ROI Analysis

EXECUTIVE SUMMARY & COMMERCIAL VALUE PROPOSITION

Opportunity Statement: Kenya Pipeline Company (KPC) currently experiences significant revenue leakage and operational inefficiencies caused by manual demurrage calculations, 45-day dispute settlement cycles, and uncaptured truck loading yard dwell-time across gantries. The **KPC Revenue Assurance & Demurrage Platform** modernizes this workflow by converting gate telemetry timestamps into deterministic, real-time invoices integrated with SAP S/4HANA. Deployment achieves a **0.35-year payback period (4.2 months)** and a **3-Year Net ROI of 761%**, transitioning demurrage management from a cost-center bottleneck into a high-yield revenue stream.

KES 35.0M	KES 100.5M	4.2 Months	761.4%	KES 215.0M
Capital Investment (CAPEX)	Net Annual Savings	Payback Period	3-Year Net ROI	3-Yr Net Present Value

1. Financial Justification & Revenue Formulas

Across KPC's primary depot gantries (Nairobi, Eldoret, Kisumu, Nakuru), an estimated 120,000 truck movements occur annually. Under current manual procedures, over 60% of dwell-time violations are either unbilled or tied up in protracted 45-day disputes.

Financial Component	Transparent Baseline Formula	Annual Impact (KES)
Gross Demurrage Recovery	15,000 excess dwell events/yr × KES 12,000 avg fee × (92% automated capture - 40% manual baseline capture)	+ KES 93,600,000
Dispute & Legal Cost Reduction	Elimination of 45-day manual dispute cycles, legal arbitration, and staff dispute resolution hours	+ KES 14,400,000
Annual Operating Costs (OPEX)	Cloud infrastructure, security maintenance, SLA support, and system license costs	- KES 7,500,000
Net Annual Financial Benefit	Net Revenue Recovery + Dispute Savings - Annual OPEX	= KES 100,500,000 / yr

2. Payback Timeline & Return on Investment (ROI) Analysis

Metric	Formula / Methodology	Quantified Value
CAPEX (Initial Investment)	Software build, SAP integration adapters, hardware gate telemetry setup	KES 35,000,000
Payback Period	CAPEX ÷ Net Annual Benefit (KES 35.0M ÷ KES 100.5M)	0.35 Years (~4.2 Months)
3-Year Net Cash Flow	(3 Years × KES 100.5M Annual Net Benefit) - KES 35.0M CAPEX	KES 266,500,000
3-Year Return on Investment	[(Total 3-Yr Net Benefit - CAPEX) ÷ CAPEX] × 100	761.4% ROI
Net Present Value (NPV)	Discounted at 10% Hurdle Rate over 3-Year Lifecycle	KES 214,883,546

3. Operational Impact & Strategic Architecture

- **Real-Time Gate Telemetry:** Captures automated entry/exit logs (gate_in / gate_out) replacing manual paper logbooks and eliminating human calculation error.
- **Automated SAP S/4HANA Ledger Posting:** Generates compliant OData billing payloads and pushes verified demurrage charges directly to KPC's enterprise ERP, streamlining billing cycles.
- **Immutable Audit Compliance:** Enforces DB-level tamper-proof append-only triggers, giving Senior Auditors and OMC representatives (e.g. Vivo Energy) 100% verifiable data transparency.

4. Strategic Risk Management & Mitigation Matrix

Identified Strategic Risk	Severity / Likelihood	Mitigation Strategy & Architecture Control
Oil Marketing Company (OMC) Resistance & Disputes	Medium / Medium	Provide self-service OMC portal access with raw telemetry evidence, automated rule verification, and clear audit trails.
Gate Telemetry Hardware or Network Outages	High / Low	Edge offline buffering and resilient CSV upload ingestion routines that auto-reconcile and recalculate upon network restoration.
Legacy ERP / SAP Integration Friction	Medium / Low	Decoupled asynchronous REST API adapter architecture with staging queues to prevent billing blocking during SAP maintenance.

STRATEGIC RECOMMENDATION FOR THE CFO & BOARD

Board Action Requested: Approval of **KES 35,000,000 CAPEX** for full-scale deployment across KPC gantries. With full payback achieved within **4.2 months** and **KES 100.5M in annual net recurring savings**, the platform is commercially compelling, operationally vital, and strategically low-risk.